

## Mark Scheme

Q1.

| Question | Scheme                                                                                  | Marks | AOs  |
|----------|-----------------------------------------------------------------------------------------|-------|------|
| (a)      | $y = x^3 - 10x^2 + 27x - 23 \Rightarrow \frac{dy}{dx} = 3x^2 - 20x + 27$                | B1    | 1.1b |
|          | $\left(\frac{dy}{dx}\right)_{x=5} = 3 \times 5^2 - 20 \times 5 + 27 (= 2)$              | M1    | 1.1b |
|          | $y + 13 = 2(x - 5)$                                                                     | M1    | 2.1  |
|          | $y = 2x - 23$                                                                           | A1    | 1.1b |
|          |                                                                                         | (4)   |      |
| (b)      | Both $C$ and $l$ pass through $(0, -23)$<br>and so $C$ meets $l$ again on the $y$ -axis | B1    | 2.2a |
|          |                                                                                         | (1)   |      |

|                    |                                                                                                                                               |            |              |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------|
| (c)                | $\pm \int (x^3 - 10x^2 + 27x - 23 - (2x - 23)) dx$ $= \pm \left( \frac{x^4}{4} - \frac{10}{3}x^3 + \frac{25}{2}x^2 \right)$                   | M1<br>A1ft | 1.1b<br>1.1b |
|                    | $\left[ \frac{x^4}{4} - \frac{10}{3}x^3 + \frac{25}{2}x^2 \right]_0^5$ $= \left( \frac{625}{4} - \frac{1250}{3} + \frac{625}{2} \right) (-0)$ | dM1        | 2.1          |
|                    | $= \frac{625}{12}$                                                                                                                            | A1         | 1.1b         |
|                    |                                                                                                                                               | (4)        |              |
| (c) Alternative 1: |                                                                                                                                               |            |              |
|                    | $\pm \int (x^3 - 10x^2 + 27x - 23) dx$ $= \pm \left( \frac{x^4}{4} - \frac{10}{3}x^3 + \frac{27}{2}x^2 - 23x \right)$                         | M1<br>A1   | 1.1b<br>1.1b |
|                    | $\left[ \frac{x^4}{4} - \frac{10}{3}x^3 + \frac{27}{2}x^2 - 23x \right]_0^5 + \frac{1}{2} \times 5(23+13)$ $= -\frac{455}{12} + 90$           | dM1        | 2.1          |
|                    | $= \frac{625}{12}$                                                                                                                            | A1         | 1.1b         |
|                    |                                                                                                                                               |            |              |
| (c) Alternative 2: |                                                                                                                                               |            |              |
|                    | $\int (x^3 - 10x^2 + 27x) dx = \left( \frac{x^4}{4} - \frac{10}{3}x^3 + \frac{27}{2}x^2 \right)$                                              | M1<br>A1   | 1.1b<br>1.1b |
|                    | $\left[ \frac{x^4}{4} - \frac{10}{3}x^3 + \frac{27}{2}x^2 \right]_0^5 - \frac{1}{2} \times 5 \times 10$                                       | dM1        | 2.1          |
|                    | $= \frac{625}{12}$                                                                                                                            | A1         | 1.1b         |
|                    |                                                                                                                                               |            |              |
| (9 marks)          |                                                                                                                                               |            |              |

### Notes

(a)

B1: Correct derivative

M1: Substitutes  $x = 5$  into their derivative. This may be implied by their value for  $\frac{dy}{dx}$

M1: Fully correct straight line method using  $(5, -13)$  and their  $\frac{dy}{dx}$  at  $x = 5$

A1: cao. Must see the full equation in the required form.

(b)

B1: Makes a suitable deduction.

Alternative via equating  $l$  and  $C$  and factorising e.g.

$$x^3 - 10x^2 + 27x - 23 = 2x - 23$$

$$x^3 - 10x^2 + 25x = 0$$

$$x(x^2 - 10x + 25) = 0 \Rightarrow x = 0$$

So they meet on the  $y$ -axis

(c)

M1: For an attempt to integrate  $x^n \rightarrow x^{n+1}$  for  $\pm C - l$

A1ft: Correct integration in any form which may be simplified or unsimplified. (follow through their equation from (a))

If they attempt as 2 separate integrals e.g.  $\int (x^3 - 10x^2 + 27x - 23) dx - \int (2x - 23) dx$  then

award this mark for the correct integration of the curve as in the alternative.

If they combine the curve with the line first then the subsequent integration must be correct or a correct ft for their line and allow for  $\pm C - l$

dM1: Fully correct strategy for the area. Award for use of 5 as the limit and condone the omission of the  $- 0$ . **Depends on the first method mark.**

A1: Correct exact value

**Alternative 1:**

M1: For an attempt to integrate  $x^n \rightarrow x^{n+1}$  for  $\pm C$

A1: Correct integration for  $\pm C$

dM1: Fully correct strategy for the area e.g. correctly attempts the area of the trapezium and subtracts the area enclosed between the curve and the  $x$ -axis. Need to see the use of 5 as the limit condoning the omission of the  $- 0$  and a correct attempt at the trapezium and the subtraction.

May see the trapezium area attempted as  $\int (2x - 23) dx$  in which case the integration and

use of the limits needs to be correct or correct follow through for their straight line equation.

**Depends on the first method mark.**

A1: Correct exact value

Note if they do  $l - C$  rather than  $C - l$  and the working is otherwise correct allow full marks if their final answer is given as a positive value. E.g. correct work with  $l - C$  leading to  $-\frac{625}{12}$  and

then e.g. hence area is  $\frac{625}{12}$  is acceptable for full marks.

If the answer is left as  $-\frac{625}{12}$  then score A0

**Alternative 2:**

M1: For an attempt to integrate  $x^n \rightarrow x^{n+1}$  for  $(C + 23)$

A1: Correct integration for  $(C + 23)$

dM1: Fully correct strategy for the area e.g. correctly attempts the area of the triangle and subtracts from the area under the curve

Need to see the use of 5 as the limit condoning the omission of the  $- 0$  and a correct attempt at the triangle and the subtraction.

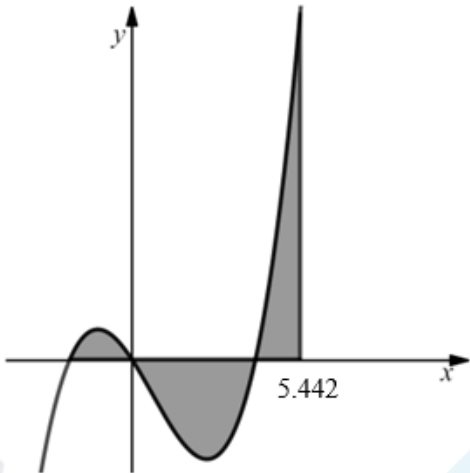
**Depends on the first method mark.**

Q2.

| Part                   | Working or answer an examiner might expect to see                                                                          | Mark | Notes                                                                  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------|
|                        | $\lim_{\delta x \rightarrow 0} \sum_{x=4}^9 \sqrt{x} \delta x = \int_4^9 \sqrt{x} \, dx$                                   | B1   | This mark is given for writing the expression for a sum as an integral |
|                        | $\left[ \frac{2}{3} x^{\frac{3}{2}} \right]_4^9 = \frac{2}{3} \times 9^{\frac{3}{2}} - \frac{2}{3} \times 4^{\frac{3}{2}}$ | M1   | This mark is given for a method to evaluate the integral               |
|                        | $= \frac{38}{3}$                                                                                                           | A1   | This mark is given for a correct evaluation of the integral            |
| <b>(Total 3 marks)</b> |                                                                                                                            |      |                                                                        |

Q3.

| Part | Working or answer an examiner might expect to see                               | Mark | Notes                                                                   |
|------|---------------------------------------------------------------------------------|------|-------------------------------------------------------------------------|
| (a)  | $y = x(x+2)(x-4) = x^3 - 2x^2 - 8x$                                             | B1   | This mark is given for expanding brackets as a first step to a solution |
|      | $\int_{-2}^0 x^3 - 2x^2 - 8x \, dx$                                             | M1   | This mark is given for a method to find the exact area of $R_1$         |
|      | $= \left[ \frac{1}{4} x^4 - \frac{2}{3} x^3 - 4x^2 \right]_{-2}^0$              | M1   | This mark is given for a method to evaluate the integral                |
|      | $= 0 - (4 - \frac{16}{3} - 16) = \frac{20}{3}$                                  | A1   | This mark is given for a full method to show the exact value of $R_1$   |
| (b)  | $\frac{1}{4} b^4 - \frac{2}{3} b^3 - 4b^2 = -\frac{20}{3}$                      | M1   | This mark is given for deducing the area of $R_2 = -\frac{20}{3}$       |
|      | $3b^4 - 8b^3 - 48b^2 + 80 = 0$                                                  | A1   | This mark is given for rearranging the equation to a quartic            |
|      | $(b+2)^2(3b^2 - 20b + 20)$<br>$= (b^2 + 4b + 4)(3b^2 - 20b + 20)$               | M1   | This mark is given for expanding the equation given                     |
|      | $= 3b^4 - 8b^3 - 48b^2 + 80 = 0$<br>The two equations are the same, so verified | A1   | This mark is for showing, and stating, that the equations are the same  |

|     |                                                                                                             |    |                                                                                  |
|-----|-------------------------------------------------------------------------------------------------------------|----|----------------------------------------------------------------------------------|
| (c) |                            | B1 | This mark is given for a sketch of the curve with $b = 5.442$ shown              |
|     | Between $x = -2$ and $b = 5.442$ , the area above the $x$ -axis is the same as the area below the $x$ -axis | B1 | This mark is given for a valid explanation of the significance of the root 5.442 |

Q4.

| Question | Scheme                                                                                                                                                                                                                                            | Marks | AOs  |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|
|          | $C: y = x \ln x$ ; $l$ is a normal to $C$ at $P(e, e)$<br>Let $x_A$ be the $x$ -coordinate of where $l$ cuts the $x$ -axis                                                                                                                        |       |      |
|          | $\frac{dy}{dx} = \ln x + x \left( \frac{1}{x} \right) \quad \{= 1 + \ln x\}$                                                                                                                                                                      | M1    | 2.1  |
|          |                                                                                                                                                                                                                                                   | A1    | 1.1b |
|          | $x = e, m_T = 2 \Rightarrow m_N = -\frac{1}{2} \Rightarrow y - e = -\frac{1}{2}(x - e)$<br>$y = 0 \Rightarrow -e = -\frac{1}{2}(x - e) \Rightarrow x = \dots$                                                                                     | M1    | 3.1a |
|          | $l$ meets $x$ -axis at $x = 3e$ (allow $x = 2e + e \ln e$ )                                                                                                                                                                                       | A1    | 1.1b |
|          | {Areas:} either $\int_1^e x \ln x \, dx = [ \dots ]_1^e = \dots$ or $\frac{1}{2}((\text{their } x_A) - e)e$                                                                                                                                       | M1    | 2.1  |
|          | $\left\{ \int x \ln x \, dx = \right\} \frac{1}{2}x^2 \ln x - \int \frac{1}{x} \left( \frac{x^2}{2} \right) \{dx\}$                                                                                                                               | M1    | 2.1  |
|          | $\left\{ = \frac{1}{2}x^2 \ln x - \int \frac{1}{2}x \{dx\} \right\} = \frac{1}{2}x^2 \ln x - \frac{1}{4}x^2$                                                                                                                                      | dM1   | 1.1b |
|          |                                                                                                                                                                                                                                                   | A1    | 1.1b |
|          | $\text{Area}(R_1) = \int_1^e x \ln x \, dx = [ \dots ]_1^e = \dots$ ; $\text{Area}(R_2) = \frac{1}{2}((\text{their } x_A) - e)e$<br>and so, $\text{Area}(R) = \text{Area}(R_1) + \text{Area}(R_2) \quad \{= \frac{1}{4}e^2 + \frac{1}{4} + e^2\}$ | M1    | 3.1a |
|          | $\text{Area}(R) = \frac{5}{4}e^2 + \frac{1}{4}$                                                                                                                                                                                                   | A1    | 1.1b |
|          |                                                                                                                                                                                                                                                   | (10)  |      |

| Notes for Question |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1:                | Differentiates by using the product rule to give $\ln x + x(\text{their } g'(x))$ , where $g(x) = \ln x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| A1:                | Correct differentiation of $y = x \ln x$ , which can be un-simplified or simplified                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| M1:                | Complete strategy to find the $x$ coordinate where their normal to $C$ at $P(e, e)$ meets the $x$ -axis<br>i.e. Sets $y=0$ in $y-e = m_N(x-e)$ to find $x = \dots$                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Note:              | $m_T$ is found by using calculus and $m_N \neq m_T$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| A1:                | $l$ meets $x$ -axis at $x = 3e$ , allowing un-simplified values for $x$ such as $x = 2e + e \ln e$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Note:              | Allow $x = \text{awrt } 8.15$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| M1:                | Scored for either <ul style="list-style-type: none"> <li>Area under curve <math>= \int_1^e x \ln x \, dx = [ \dots ]_1^e = \dots</math>, with limits of <math>e</math> and <math>1</math> and some attempt to substitute these and subtract</li> <li>or Area under line <math>= \frac{1}{2}((\text{their } x_A) - e)e</math>, with a valid attempt to find <math>x_A</math></li> </ul>                                                                                                                                                                                                                               |
| M1:                | Integration by parts the correct way around to give $Ax^2 \ln x - \int B \left( \frac{x^2}{x} \right) \{dx\}$ ; $A \neq 0$ , $B > 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| dM1:               | dependent on the previous M mark<br>Integrates the second term to give $\pm \lambda x^2$ ; $\lambda \neq 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| A1:                | $\frac{1}{2}x^2 \ln x - \frac{1}{4}x^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| M1:                | Complete strategy of finding the area of $R$ by finding the sum of two key areas. See scheme.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| A1:                | $\frac{5}{4}e^2 + \frac{1}{4}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Note:              | Area( $R_2$ ) can also be found by integrating the line $l$ between limits of $e$ and their $x_A$<br>i.e. $\text{Area}(R_2) = \int_e^{\text{their } x_A} \left( -\frac{1}{2}x + \frac{3}{2}e \right) dx = [ \dots ]_e^{\text{their } x_A} = \dots$                                                                                                                                                                                                                                                                                                                                                                   |
| Note:              | <b>Calculator approach with no algebra, differentiation or integration seen:</b> <ul style="list-style-type: none"> <li>Finding <math>l</math> cuts through the <math>x</math>-axis at awrt 8.15 is 2<sup>nd</sup> M1 2<sup>nd</sup> A1</li> <li>Finding area between curve and the <math>x</math>-axis between <math>x=1</math> and <math>x=e</math> to give awrt 2.10 is 3<sup>rd</sup> M1</li> <li>Using the above information (must be seen) to apply<br/> <math>\text{Area}(R) = 2.0972\dots + 7.3890\dots = 9.4862\dots</math> is final M1</li> </ul> Therefore, a maximum of 4 marks out of the 10 available. |



Q5.

| Question         | Scheme                                                                                                                                                                                                                                                   | Marks     | AOs          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------|
|                  | $y = \frac{(x-2)(x-4)}{4\sqrt{x}} = \frac{x^2 - 6x + 8}{4\sqrt{x}} = \frac{1}{4}x^{\frac{3}{2}} - \frac{3}{2}x^{\frac{1}{2}} + 2x^{-\frac{1}{2}}$                                                                                                        | M1<br>A1  | 1.1b<br>1.1b |
|                  | $\int \frac{1}{4}x^{\frac{3}{2}} - \frac{3}{2}x^{\frac{1}{2}} + 2x^{-\frac{1}{2}} dx = \frac{1}{10}x^{\frac{5}{2}} - x^{\frac{3}{2}} + 4x^{\frac{1}{2}} (+c)$                                                                                            | dM1<br>A1 | 3.1a<br>1.1b |
|                  | Deduces limits of integral are 2 and 4 and applies to their<br>$\frac{1}{10}x^{\frac{5}{2}} - x^{\frac{3}{2}} + 4x^{\frac{1}{2}}$                                                                                                                        | M1        | 2.2a         |
|                  | $\left(\frac{32}{10} - 8 + 8\right) - \left(\frac{2}{5}\sqrt{2} - 2\sqrt{2} + 4\sqrt{2}\right) = \frac{16}{5} - \frac{12}{5}\sqrt{2}$ $\text{Area } R = \frac{12}{5}\sqrt{2} - \frac{16}{5} \left(\text{or } \frac{16}{5} - \frac{12}{5}\sqrt{2}\right)$ | A1        | 2.1          |
|                  |                                                                                                                                                                                                                                                          | (6)       |              |
| <b>(6 marks)</b> |                                                                                                                                                                                                                                                          |           |              |
| <b>Notes:</b>    |                                                                                                                                                                                                                                                          |           |              |

**M1:** Correct attempt to write  $\frac{(x-2)(x-4)}{4\sqrt{x}}$  as a sum of terms with indices.

Look for at least two different terms with the correct index e.g. two of  $x^{\frac{3}{2}}$ ,  $x^{\frac{1}{2}}$ ,  $x^{-\frac{1}{2}}$  which have come from the correct places.

The correct indices may be implied later when e.g.  $\sqrt{x}$  becomes  $x^{\frac{1}{2}}$  or  $\frac{1}{\sqrt{x}}$  becomes  $x^{-\frac{1}{2}}$

**A1:**  $\frac{1}{4}x^{\frac{3}{2}} - \frac{3}{2}x^{\frac{1}{2}} + 2x^{-\frac{1}{2}}$  which can be left unsimplified e.g.  $\frac{1}{4}x^{2-\frac{1}{2}} - \frac{1}{2}x^{\frac{1}{2}} - x^{\frac{1}{2}} + 2x^{-\frac{1}{2}}$   
 or as e.g.  $\frac{1}{4}\left(x^{\frac{3}{2}} - 6x^{\frac{1}{2}} + 8x^{-\frac{1}{2}}\right)$

The correct indices may be implied later when e.g.  $\sqrt{x}$  becomes  $x^{\frac{1}{2}}$  or  $\frac{1}{\sqrt{x}}$  becomes  $x^{-\frac{1}{2}}$

**dM1:** Integrates  $x^n \rightarrow x^{n+1}$  for at least 2 correct indices

i.e. at least 2 of  $x^{\frac{3}{2}} \rightarrow x^{\frac{5}{2}}$ ,  $x^{\frac{1}{2}} \rightarrow x^{\frac{3}{2}}$ ,  $x^{-\frac{1}{2}} \rightarrow x^{\frac{1}{2}}$

It is dependent upon the first M so at least two terms must have had a correct index.

**A1:**  $\frac{1}{10}x^{\frac{5}{2}} - x^{\frac{3}{2}} + 4x^{\frac{1}{2}} (+c)$ . Allow unsimplified e.g.  $\frac{1}{4} \times \frac{2}{5}x^{\frac{3}{2}+1} - \frac{1}{2} \times \frac{2}{3}x^{\frac{1}{2}+1} - \frac{2}{3}x^{\frac{1}{2}+1} + 2 \times 2x^{\frac{1}{2}}$   
 or as e.g.  $\frac{1}{4}\left(\frac{2}{5}x^{\frac{5}{2}} - 4x^{\frac{3}{2}} + 16x^{\frac{1}{2}}\right) (+c)$ .

**M1:** Substitutes the limits 4 and 2 to their  $\frac{1}{10}x^{\frac{5}{2}} - x^{\frac{3}{2}} + 4x^{\frac{1}{2}}$  and subtracts either way round.

There is no requirement to evaluate but 4 and 2 must be substituted either way round with evidence of subtraction, condoning omission of brackets.

E.g. condone  $\frac{1}{10} \times 4^{\frac{5}{2}} - 4^{\frac{3}{2}} + 4 \times 4^{\frac{1}{2}} - \frac{1}{10} \times 2^{\frac{5}{2}} - 2^{\frac{3}{2}} + 4 \times 2^{\frac{1}{2}}$

This is an independent mark but the limits must be applied to an expression that is not  $y$  so they may even have differentiated.

**A1:** Correct working shown leading to  $\frac{12}{5}\sqrt{2} - \frac{16}{5}$  but also allow  $\frac{16}{5} - \frac{12}{5}\sqrt{2}$  or exact equivalents

Award this mark once one of these forms is reached and isw



## Integration by parts:

|                                                                                                                                                                                                                                                                                                                                                                                                    |           |              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------|
| $\int \frac{(x-2)(x-4)}{4\sqrt{x}} dx = \int \frac{1}{4}(x-2)(x-4)x^{-\frac{1}{2}} dx = \frac{1}{2}(x-2)(x-4)x^{\frac{1}{2}} - \int \frac{1}{2}(2x-6)x^{\frac{1}{2}} dx$                                                                                                                                                                                                                           | M1<br>A1  | 1.1b<br>1.1b |
| $\frac{1}{2}(x-2)(x-4)x^{\frac{1}{2}} - \int \frac{1}{2}(2x-6)x^{\frac{1}{2}} dx = \frac{1}{2}(x-2)(x-4)x^{\frac{1}{2}} - \int x^{\frac{3}{2}} - 3x^{\frac{1}{2}} dx$ $= \frac{1}{2}(x-2)(x-4)x^{\frac{1}{2}} - \frac{2}{5}x^{\frac{5}{2}} + 2x^{\frac{3}{2}}$ <p>Or e.g. <math>= \frac{1}{2}(x-2)(x-4)x^{\frac{1}{2}} - \frac{1}{3}x^{\frac{3}{2}}(2x-6) + \frac{4}{15}x^{\frac{5}{2}}</math></p> | dM1<br>A1 | 3.1a<br>1.1b |
| Deduces limits of integral are 2 and 4 and applies to their<br>$\frac{1}{2}(x-2)(x-4)x^{\frac{1}{2}} - \frac{1}{3}x^{\frac{3}{2}}(2x-6) + \frac{4}{15}x^{\frac{5}{2}}$                                                                                                                                                                                                                             | M1        | 2.2a         |
| $0 - \frac{16}{3} + \frac{128}{15} - \left( 0 + \frac{4}{3}\sqrt{2} + \frac{16}{15}\sqrt{2} \right)$ $\text{Area } R = \frac{12}{5}\sqrt{2} - \frac{16}{5} \left( \text{or } \frac{16}{5} - \frac{12}{5}\sqrt{2} \right)$                                                                                                                                                                          | A1        | 2.1          |
|                                                                                                                                                                                                                                                                                                                                                                                                    | (6)       |              |

Notes:

M1: Applies integration by parts and reaches the form  $\alpha(x-2)(x-4)x^{\frac{1}{2}} \pm \int (px+q)x^{\frac{1}{2}} dx$   $\alpha, p \neq 0$

oe e.g.  $\alpha(x^2 - 6x + 8)x^{\frac{1}{2}} \pm \int (px+q)x^{\frac{1}{2}} dx$   $\alpha, p \neq 0$

A1: Correct first application of parts in any form

dM1: Attempts their  $\int (px+q)x^{\frac{1}{2}} dx$  by expanding and integrating or may attempt parts again.

E.g.  $\int (2x-6)x^{\frac{1}{2}} dx = \int \left( 2x^{\frac{3}{2}} - 6x^{\frac{1}{2}} \right) dx = \dots$  or e.g.  $\int (2x-6)x^{\frac{1}{2}} dx = \frac{2}{3}x^{\frac{3}{2}}(2x-6) - \frac{4}{3} \int x^{\frac{3}{2}} dx$

If they expand then at least one term requires  $x^n \rightarrow x^{n+1}$  or if parts is attempted again, the structure must be correct.

A1: Fully correct integration in any form

M1: Substitutes the limits 4 and 2 to their  $= \frac{1}{2}(x-2)(x-4)x^{\frac{1}{2}} - \frac{2}{5}x^{\frac{5}{2}} + 2x^{\frac{3}{2}}$  and subtracts

either way round. There is no requirement to evaluate but 4 and 2 must be substituted either way round with evidence of subtraction, condoning omission of brackets.

E.g. condone  $0 - \frac{16}{3} + \frac{128}{15} - 0 + \frac{4}{3}\sqrt{2} + \frac{16}{15}\sqrt{2}$

This is an independent mark but the limits must be applied to a "changed" function.

A1: Correct working shown leading to  $\frac{12}{5}\sqrt{2} - \frac{16}{5}$  but also allow  $\frac{16}{5} - \frac{12}{5}\sqrt{2}$  or exact equivalents

Attempts at integration by parts "the other way round" should be sent to review.





## Integration by substitution example:

|                                                                                                                                                                                                                                       |           |              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------|
| $u = \sqrt{x} \ (x = u^2) \Rightarrow \int \frac{(x-2)(x-4)}{4\sqrt{x}} dx = \int \frac{(u^2-2)(u^2-4)}{4u} \frac{dx}{du} du$ $= \int \frac{(u^2-2)(u^2-4)}{4u} 2u du$                                                                | M1<br>A1  | 1.1b<br>1.1b |
| $= \frac{1}{2} \int (u^4 - 6u^2 + 8) du = \frac{1}{2} \left( \frac{u^5}{5} - \frac{6u^3}{3} + 8u \right) (+c)$                                                                                                                        | dM1<br>A1 | 3.1a<br>1.1b |
| Deduces limits of integral are $\sqrt{2}$ and 2 and applies to their<br>$\frac{1}{2} \left( \frac{u^5}{5} - \frac{6u^3}{3} + 8u \right)$                                                                                              | M1        | 2.2a         |
| $\frac{1}{2} \left( \frac{32}{5} - 16 + 16 - \left( \frac{4\sqrt{2}}{5} - 4\sqrt{2} + 8\sqrt{2} \right) \right)$ $\text{Area } R = \frac{12}{5}\sqrt{2} - \frac{16}{5} \left( \text{or } \frac{16}{5} - \frac{12}{5}\sqrt{2} \right)$ | A1        | 2.1          |
|                                                                                                                                                                                                                                       | (6)       |              |

## Notes:

M1: Applies the substitution e.g.  $u = \sqrt{x}$  and attempts  $k \int \frac{(u^2-2)(u^2-4)}{u} \frac{dx}{du} du$

A1: Fully correct integral in terms of  $u$  in any form e.g.  $\frac{1}{2} \int (u^2-2)(u^2-4) du$

dM1: Expands the bracket and integrates  $u^n \rightarrow u^{n+1}$  for at least 2 correct indices

i.e. at least 2 of  $u^4 \rightarrow u^5$ ,  $u^2 \rightarrow u^3$ ,  $k \rightarrow ku$

A1:  $\frac{1}{2} \left( \frac{u^5}{5} - \frac{6u^3}{3} + 8u \right) (+c)$ . Allow unsimplified.

M1: Substitutes the limits 2 and  $\sqrt{2}$  to their  $\frac{1}{2} \left( \frac{u^5}{5} - \frac{6u^3}{3} + 8u \right)$  and subtracts either way round.

There is no requirement to evaluate but 2 and  $\sqrt{2}$  must be substituted either way round with evidence of subtraction, condoning omission of brackets.

E.g. condone  $\frac{1}{2} \left( \frac{32}{5} - 16 + 16 - \frac{4\sqrt{2}}{5} - 4\sqrt{2} + 8\sqrt{2} \right)$

Alternatively reverses the substitution and applies the limits 4 and 2 with the same conditions.

A1: Correct working shown leading to  $\frac{12}{5}\sqrt{2} - \frac{16}{5}$  but also allow  $\frac{16}{5} - \frac{12}{5}\sqrt{2}$  or exact equivalents

Award this mark once one of these forms is reached and isw.

There may be other substitutions seen and the same marking principles apply.

| Part | Working or answer an examiner might expect to see                        | Mark | Notes                                                                                         |
|------|--------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------|
| (a)  | The asymptote is found where $2x - q = 0$<br>Hence $q = 4$               | B1   | This mark is given for explaining that the asymptote at $x = 2$ is a solution of $2x - q = 0$ |
|      | $y = \frac{p-3x}{(2x-4)(x+3)}$<br>$\frac{1}{2} = \frac{p-9}{(6-4)(3+3)}$ | M1   | This mark is given for substituting $x = 3$ ,<br>$y = \frac{1}{2}$ (and $q = 4$ )             |

|     |                                                                                                                                                                                      |    |                                                                                               |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----------------------------------------------------------------------------------------------|
| (b) | $\frac{15-3x}{(2x-4)(x+3)} = \frac{A}{(2x-4)} + \frac{B}{(x+3)}$                                                                                                                     | M1 | This mark is given for a method to use partial fractions                                      |
|     | $= \frac{1.8}{(2x-4)} - \frac{2.4}{(x+3)}$                                                                                                                                           | M1 | This mark is given for finding values for $A$ and $B$                                         |
|     | $= \frac{0.9}{(x-2)} - \frac{2.4}{(x+3)}$                                                                                                                                            | A1 | This mark is given for a fully simplified expression                                          |
|     | $I = \int \frac{15-3x}{(2x-4)(x+3)} dx$<br>$= m \ln(2x-4) + n \ln(x+3)$                                                                                                              | M1 | This mark is given for a method to integrate to find the area of $R$                          |
|     | $= 0.9 \ln(2x-4) + 2.4 \ln(x+3)$                                                                                                                                                     | A1 | This mark is given for a correct expression for the area of $R$                               |
|     | $\text{Area } R = \left[ 0.9 \ln(2x-4) - 2.4 \ln(x+3) \right]_3^5$                                                                                                                   | M1 | This mark is given for deducing an expression for the area of $R$<br>( $y = 0$ when $x = 5$ ) |
|     | $= [0.9 \ln 6 - 2.4 \ln 8] - [0.9 \ln 2 - 2.4 \ln 6]$<br>$= [0.9 \ln 6 + 2.4 \ln 6] - [7.2 \ln 2 + 0.9 \ln 2]$<br>$= 3.3 \ln 6 - 8.1 \ln 2$<br>$= 3.3 \ln 3 + 3.3 \ln 2 - 8.1 \ln 2$ | M1 | This mark is given for a method to find the exact area of $R$                                 |
|     | $= 3.3 \ln 3 - 4.8 \ln 2$                                                                                                                                                            | A1 | This mark is given for a correct value of the area of $R$ with $a = 3.3$ and $b = 4.8$        |

Q7.

**Via firstly integrating**

| Question         | Scheme                                                                                                                     | Marks    | AOs          |
|------------------|----------------------------------------------------------------------------------------------------------------------------|----------|--------------|
|                  | $f'(x) = 6x^2 + ax - 23 \Rightarrow f(x) = 2x^3 + \frac{1}{2}ax^2 - 23x + c$                                               | M1<br>A1 | 1.1b<br>1.1b |
|                  | "c" = -12                                                                                                                  | B1       | 2.2a         |
|                  | $f(-4) = 0 \Rightarrow 2 \times (-4)^3 + \frac{1}{2}a(-4)^2 - 23(-4) - 12 = 0$                                             | dM1      | 3.1a         |
|                  | $a = \dots (6)$                                                                                                            | dM1      | 1.1b         |
|                  | $(f(x) =) 2x^3 + 3x^2 - 23x - 12$<br>Or Equivalent e.g.<br>$(f(x) =) (x+4)(2x^2 - 5x - 3) \quad (f(x) =) (x+4)(2x+1)(x-3)$ | A1cso    | 2.1          |
|                  |                                                                                                                            | (6)      |              |
| <b>(6 marks)</b> |                                                                                                                            |          |              |

**Notes:**

**M1:** Integrates  $f'(x)$  with two correct indices. There is no requirement for the + c

**A1:** Fully correct integration (may be unsimplified). The + c must be seen (or implied by the -12)

**B1:** Deduces that the constant term is -12

**dM1:** Dependent upon having done some integration. It is for setting up a linear equation in a by using  $f(-4) = 0$

May also see long division attempted for this mark. Need to see a complete method leading to a remainder in terms of a which is then set = 0.

For reference, the quotient is  $2x^2 + \left(\frac{a}{2} - 8\right)x + 9 - 2a$  and the remainder is  $8a - 48$

May also use  $(x+4)(px^2 + qx + r) = 2x^3 + \frac{1}{2}ax^2 - 23x - 12$  and compare coefficients to find p, q and r and hence a. Allow this mark if they solve for p, q and r

Note that some candidates use  $2f(x)$  which is acceptable and gives the same result if executed correctly.

**dM1:** Solves the linear equation in a or uses p, q and r to find a.

It is dependent upon having attempted some integration and used  $f(\pm 4) = 0$  or long division/comparing coefficients with  $(x+4)$  as a factor.

**A1cso:** For  $(f(x) =) 2x^3 + 3x^2 - 23x - 12$  oe. Note that "f(x) =" does not need to be seen and ignore any "= 0"

**Via firstly using factor**

| Question         | Scheme                                                                                      | Marks    | AOs          |
|------------------|---------------------------------------------------------------------------------------------|----------|--------------|
| <b>Alt</b>       | $f(x) = (x+4)(Ax^2 + Bx + C)$                                                               | M1<br>A1 | 1.1b<br>1.1b |
|                  | $f(x) = Ax^3 + (4A+B)x^2 + (4B+C)x + 4C \Rightarrow C = -3$                                 | B1       | 2.2a         |
|                  | $f'(x) = 3Ax^2 + 2(4A+B)x + (4B+C)$ and $f'(x) = 6x^2 + ax - 23$<br>$\Rightarrow A = \dots$ | dM1      | 3.1a         |
|                  | Full method to get A, B and C                                                               | dM1      | 1.1b         |
|                  | $f(x) = (x+4)(2x^2 - 5x - 3)$                                                               | A1cso    | 2.1          |
|                  |                                                                                             | (6)      |              |
| <b>(6 marks)</b> |                                                                                             |          |              |

**Notes:**

**M1:** Uses the fact that  $f(x)$  is a cubic expression with a factor of  $(x+4)$

**A1:** For  $f(x) = (x+4)(Ax^2 + Bx + C)$

**B1:** Deduces that  $C = -3$

**dM1:** Attempts to differentiate either by product rule or via multiplication and compares to  $f'(x) = 6x^2 + ax - 23$  to find A.

**dM1:** Full method to get A, B and C

**A1cso:**  $f(x) = (x+4)(2x^2 - 5x - 3)$  or  $f(x) = (x+4)(2x+1)(x-3)$

Q8.

| Question  | Scheme                                                                                                                                                                                                                                                              | Marks | AOs  |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|
| (a)       | $x^2 + 8x - 3 = (Ax + B)(x + 2) + C$ or $Ax(x + 2) + B(x + 2) + C$<br>$\Rightarrow A = \dots, B = \dots, C = \dots$<br>or<br>$\begin{array}{r} x+6 \\ x+2 \overline{) x^2+8x-3} \\ \underline{x^2+2x} \phantom{-3} \\ 6x-3 \\ \underline{6x+12} \\ -15 \end{array}$ | M1    | 1.1b |
|           | Two of $A = 1, B = 6, C = -15$                                                                                                                                                                                                                                      | A1    | 1.1b |
|           | All three of $A = 1, B = 6, C = -15$                                                                                                                                                                                                                                | A1    | 1.1b |
|           | (3)                                                                                                                                                                                                                                                                 |       |      |
| 6(b)      | $\int \frac{x^2 + 8x - 3}{x + 2} dx = \int x + 6 - \frac{15}{x + 2} dx = \dots - 15 \ln(x + 2)$                                                                                                                                                                     | M1    | 1.1b |
|           | $= \frac{1}{2}x^2 + 6x - 15 \ln(x + 2) + c$                                                                                                                                                                                                                         | A1ft  | 1.1b |
|           | $\int_0^6 \frac{x^2 + 8x - 3}{x + 2} dx = \left[ \frac{1}{2}x^2 + 6x - 15 \ln(x + 2) \right]_0^6$<br>$= (18 + 36 - 15 \ln 8) - (0 + 0 - 15 \ln 2)$<br>$= 18 + 36 - (15 - 45) \ln 2$ or e.g. $18 + 36 + 15 \ln \left( \frac{2}{8} \right)$                           | M1    | 2.1  |
|           | $= 54 - 30 \ln 2$                                                                                                                                                                                                                                                   | A1    | 1.1b |
|           | (4)                                                                                                                                                                                                                                                                 |       |      |
| (7 marks) |                                                                                                                                                                                                                                                                     |       |      |

Notes:

- (a)
- M1:** Multiplies by  $(x + 2)$  and attempts to find values for  $A, B$  and  $C$  e.g. by comparing coefficients or substituting values for  $x$ . If the method is unclear, at least 2 terms must be correct on rhs.
- Or** attempts to divide  $x^2 + 8x - 3$  by  $x + 2$  and obtains a linear quotient and a constant remainder.
- This mark may be implied by 2 correct values for  $A, B$  or  $C$
- A1:** Two of  $A = 1, B = 6, C = -15$ . But note that just performing the division correctly is insufficient and they must clearly identify their  $A, B, C$  to score any accuracy marks.
- A1:** All three of  $A = 1, B = 6, C = -15$
- This is implied by stating  $\frac{x^2 + 8x - 3}{x + 2} = x + 6 - \frac{15}{x + 2}$  or within the integral in (b)
- (b)
- M1:** Integrates an expression of the form  $\frac{C}{x + 2}$  to obtain  $k \ln(x + 2)$ .
- Condone the omission of brackets around the " $x + 2$ "
- A1ft:** Correct integration ft on their  $Ax + B + \frac{C}{x + 2}$ , ( $A, B, C \neq 0$ ) The brackets should be present around the " $x + 2$ " unless they are implied by subsequent work.
- M1:** Substitutes both limits 0 and 6 into an expression that contains an  $x$  or  $x^2$  term or both and a  $\ln$  term and subtracts either way round **WITH** fully correct log work to combine two log terms (but allow sign errors when removing brackets) leading to an answer of the form  $a + b \ln c$  ( $a, b$  and  $c$  not necessarily integers)
- e.g. if they expand to get  $-15 \ln 8 - 15 \ln 2$  followed by  $-15 \ln 16$  and reach  $a + b \ln c$  then allow the M mark
- A1:**  $54 - 30 \ln 2$  (Apply isw once a correct answer is seen)

Q9.

| Question  | Scheme                                                                                        | Marks    | AOs          |
|-----------|-----------------------------------------------------------------------------------------------|----------|--------------|
| (a)       | $h = 0.5$                                                                                     | B1       | 1.1b         |
|           | $A \approx \frac{1}{2} \times \frac{1}{2} \{0.4805 + 1.9218 + 2(0.8396 + 1.2069 + 1.5694)\}$  | M1       | 1.1b         |
|           | $= 2.41$                                                                                      | A1       | 1.1b         |
|           |                                                                                               | (3)      |              |
| (b)       | $\int (\ln x)^2 dx = x(\ln x)^2 - \int x \times \frac{2 \ln x}{x} dx$                         | M1<br>A1 | 3.1a<br>1.1b |
|           | $= x(\ln x)^2 - 2 \int \ln x dx = x(\ln x)^2 - 2(x \ln x - \int dx)$                          | dM1      | 2.1          |
|           | $= x(\ln x)^2 - 2 \int \ln x dx = x(\ln x)^2 - 2x \ln x + 2x$                                 |          |              |
|           | $\int_2^4 (\ln x)^2 dx = \left[ x(\ln x)^2 - 2x \ln x + 2x \right]_2^4$                       | ddM1     | 2.1          |
|           | $= 4(\ln 4)^2 - 2 \times 4 \ln 4 + 2 \times 4 - (2(\ln 2)^2 - 2 \times 2 \ln 2 + 2 \times 2)$ |          |              |
|           | $= 4(2 \ln 2)^2 - 16 \ln 2 + 8 - 2(\ln 2)^2 + 4 \ln 2 - 4$                                    |          |              |
|           | $= 14(\ln 2)^2 - 12 \ln 2 + 4$                                                                | A1       | 1.1b         |
|           |                                                                                               | (5)      |              |
| (8 marks) |                                                                                               |          |              |
| Notes     |                                                                                               |          |              |



(a)

B1: Correct strip width. May be implied by  $\frac{1}{2} \times \frac{1}{2} \{ \dots \}$  or  $\frac{1}{4} \times \{ \dots \}$

M1: Correct application of the trapezium rule.

Look for  $\frac{1}{2} \times h \{ 0.4805 + 1.9218 + 2(0.8396 + 1.2069 + 1.5694) \}$  condoning slips in the digits.

The bracketing must be correct but it is implied by awrt 2.41

A1: 2.41 only. This is not awrt

(b)

M1: Attempts parts the correct way round to achieve  $\alpha x (\ln x)^2 - \beta \int \ln x \, dx$  o.e.

May be unsimplified (see scheme).

Watch for candidates who know or learn  $\int \ln x \, dx = x \ln x - x$

who may write  $\int (\ln x)^2 \, dx = \int (\ln x)(\ln x) \, dx = \ln x (x \ln x - x) - \int \frac{x \ln x - x}{x} \, dx$

A1: Correct expression which may be unsimplified

dM1: Attempts parts again to (only condone coefficient errors) to

achieve  $\alpha x (\ln x)^2 - \beta x \ln x \pm \gamma x$  o.e.

ddM1: Applies the limits 4 and 2 to an expression of the form  $\pm \alpha x (\ln x)^2 \pm \beta x \ln x \pm \gamma x$ , subtracts and applies  $\ln 4 = 2 \ln 2$  at least once. Both M's must have been awarded

A1: Correct answer

It is possible to do  $\int (\ln x)^2 \, dx$  via a substitution  $u = \ln x$  but it is very similar.

M1 A1, dM1:  $\int u^2 e^u \, du = u^2 e^u - \int 2u e^u \, du = u^2 e^u - 2u e^u \pm 2e^u$

ddM1: Applies appropriate limits and uses  $\ln 4 = 2 \ln 2$  at least once to an expression of the form  $u^2 e^u - \beta u e^u \pm \gamma e^u$  Both M's must have been awarded



Q10.

| Question | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks | AOs  |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|
| (a)      | $h = 0.5$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | B1    | 1.1a |
|          | $A \approx \frac{0.5}{2} \{0.5774 + 0.8452 + 2(0.7071 + 0.7746 + 0.8165)\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | M1    | 1.1b |
|          | = awrt 1.50                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | A1    | 1.1b |
|          | <b>For reference:</b><br>The integration on a calculator gives 1.511549071<br>The full accuracy for $y$ values gives 1.504726147<br>The accuracy from the table gives 1.50475                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |      |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | (3)   |      |
| (b)      | $3 \times \text{their (a)}$<br><br>If (a) is correct, allow awrt 4.50 or awrt 4.51 even with no working.<br>Only allow 4.5 if (a) is correct and working is shown e.g. $3 \times 1.5$<br><br>If (a) is incorrect allow $3 \times \text{their (a)}$ given to at least 3sf but do not be too concerned about the accuracy (as they may use rounded or rounded value from (a))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | B1ft  | 2.2a |
|          | For reference the integration on a calculator gives 4.534647213                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | (1)   |      |
| (c)      | <p><b>This mark depends on the B1 having been awarded in part (b) with awrt 4.5</b></p> <p>Look for a sensible comment. Some examples:</p> <ul style="list-style-type: none"> <li>The answer is accurate to 2 sf or one decimal place</li> <li>Answer to (b) is accurate as <math>4.535 \approx 4.50</math></li> <li>Very accurate as <math>4.535</math> to 2 sf is 4.5</li> <li><math>4.51425 &lt; 4.535</math> so my answer is underestimate but not too far off</li> <li>It is an underestimate but quite close</li> <li>It is a very good estimate</li> <li>High accuracy</li> <li>(Quite) accurate</li> <li>It is less than 1% out</li> <li><math>4.535 - 4.5 = 0.035</math> so not far out</li> </ul> <p>But not just "it is an underestimate"</p> <p>or</p> <p>Calculates percentage error correctly using awrt 4.50 or awrt 4.51 or 4.5<br/>           (No comment is necessary in these cases although one may be given)</p> <p>Examples:</p> $\left  \frac{4.535 - 4.50}{4.535} \right  \times 100 = 0.77\% \quad \text{or} \quad \left  \frac{4.535 - 4.51}{4.535} \right  \times 100 = 0.55\% \quad \text{or}$ $\left  \frac{4.535 - 4.51425}{4.535} \right  \times 100 = 0.46\% \quad \text{or} \quad \left  \frac{4.50}{4.535} \right  \times 100 = 99\%$ <p>In these cases don't be too concerned about accuracy e.g. allow 1sf.<br/> <b>This mark should be withheld if there are any contradictory statements</b></p> | B1    | 3.2b |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | (1)   |      |

(5 marks)

**Notes:**

(a)

**B1:** States or uses  $h = 0.5$ . May be implied by  $\frac{1}{4} \times \{ \dots \}$  below.

**M1:** Correct attempt at the trapezium rule.

Look for  $\frac{1}{2}h \times \{0.5774 + 0.8452 + 2(0.7071 + 0.7746 + 0.8165)\}$  condoning slips on the terms but must use all  $y$  values with no repeats.

There must be a clear attempt at  $\frac{1}{2}h \times (\text{first } y + \text{last } y + 2 \times \text{"sum of the rest"})$

Give M0 for  $\frac{1}{2} \times \frac{1}{2} \times 0.5774 + 0.8452 + 2(0.7071 + 0.7746 + 0.8165)$  unless the missing brackets are implied.

NB this incorrect method gives 5.85...

May be awarded for separate trapezia e.g.

$$\frac{1}{4}(0.5774 + 0.7071) + \frac{1}{4}(0.7071 + 0.7746) + \frac{1}{4}(0.7746 + 0.8165) + \frac{1}{4}(0.8165 + 0.8452)$$

May be awarded for using the function e.g.  $\frac{1}{2}h \times \left\{ \sqrt{\frac{0.5}{1+0.5}} + \sqrt{\frac{2.5}{1+2.5}} + 2 \left( \sqrt{\frac{1}{1+1}} + \sqrt{\frac{1.5}{1+1.5}} + \sqrt{\frac{2}{1+2}} \right) \right\}$

**A1:** Awrt 1.50 (Apply isw if necessary)

Correct answers with no working – send to review

(b)

**B1ft:** See main scheme. Must be considering  $3 \times$  (a) and not e.g. attempting trapezium rule again.

(c)

**B1:** See scheme

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Q11.

| Question  | Scheme                                                                                                                                                                                                                                                   | Marks    | AOs         |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|
| (a)       | Attempts<br>$y \cdot \frac{dx}{dt} = (2 \sin 2t + 3 \sin t) \times 16 \sin t \cos t$ and uses $\sin 2t = 2 \sin t \cos t$                                                                                                                                | M1       | 2.1         |
|           | Correct expanded integrand. Usually for one of<br>$(R) = \int \underline{48 \sin^2 t \cos t + 16 \sin^2 2t} dt$<br>$(R) = \int \underline{48 \sin^2 t \cos t + 64 \sin^2 t \cos^2 t} dt$<br>$(R) = \int \underline{24 \sin 2t \sin t + 16 \sin^2 2t} dt$ | A1       | 1.1b        |
|           | Attempts to use $\cos 4t = 1 - 2 \sin^2 2t = (1 - 8 \sin^2 t \cos^2 t)$                                                                                                                                                                                  | M1       | 1.1b        |
|           | $R = \int_0^a 8 - 8 \cos 4t + 48 \sin^2 t \cos t dt$ *                                                                                                                                                                                                   | A1*      | 2.1         |
|           | Deduces $a = \frac{\pi}{4}$                                                                                                                                                                                                                              | B1       | 2.2a        |
|           |                                                                                                                                                                                                                                                          | (5)      |             |
| (b)       | $\int 8 - 8 \cos 4t + 48 \sin^2 t \cos t dt = 8t - 2 \sin 4t + 16 \sin^3 t$                                                                                                                                                                              | M1<br>A1 | 2.1<br>1.1b |
|           | $\left[ 8t - 2 \sin 4t + 16 \sin^3 t \right]_0^{\frac{\pi}{4}} = 2\pi + 4\sqrt{2}$                                                                                                                                                                       | M1<br>A1 | 2.1<br>1.1b |
|           |                                                                                                                                                                                                                                                          | (4)      |             |
| (9 marks) |                                                                                                                                                                                                                                                          |          |             |
| Notes:    |                                                                                                                                                                                                                                                          |          |             |

(a) Condone work in another variable, say  $\theta \leftrightarrow t$  if used consistently for the first 3 marks

M1: For the key step in attempting  $y \cdot \frac{dx}{dt} = (2 \sin 2t + 3 \sin t) \times 16 \sin t \cos t$  with an attempt to use

$\sin 2t = 2 \sin t \cos t$  Condone slips in finding  $\frac{dx}{dt}$  but it must be of the form  $k \sin t \cos t$

E.g. I  $y \cdot \frac{dx}{dt} = (2 \sin 2t + 3 \sin t) \times k \sin t \cos t = (4 \sin t \cos t + 3 \sin t) \times k \sin t \cos t$

E.g. II  $y \cdot \frac{dx}{dt} = (2 \sin 2t + 3 \sin t) \times k \sin t \cos t = (2 \sin 2t + 3 \sin t) \times \frac{k}{2} \sin 2t$

A1: A correct (expanded) integrand in  $t$ . Don't be concerned by the absence of  $\int$  or  $dt$  or limits

$(R) = \int \underline{48 \sin^2 t \cos t + 16 \sin^2 2t} dt$  or  $(R) = \int \underline{48 \sin^2 t \cos t + 64 \sin^2 t \cos^2 t} dt$

but watch for other correct versions such as  $(R) = \int \underline{24 \sin 2t \sin t + 16 \sin^2 2t} dt$

**M1:** Attempts to use  $\cos 4t = \pm 1 \pm 2 \sin^2 2t$  to get the integrand in the correct form.

If they have the form  $P \sin^2 2t$  it is acceptable to write  $P \sin^2 2t = \frac{P}{2}(\pm 1 \pm \cos 4t)$

If they have the form  $Q \sin^2 t \cos^2 t$  sight and use of  $\sin 2t$  and/or  $\cos 2t$  will usually be seen first.

There are many ways to do this, below is such an example

$$Q \sin^2 t \cos^2 t = Q \left( \frac{1 - \cos 2t}{2} \right) \left( \frac{1 + \cos 2t}{2} \right) = Q \left( \frac{1 - \cos^2 2t}{4} \right) = Q \left( \frac{1}{4} - \frac{\cos^2 2t}{4} \right) = Q \left( \frac{1}{4} - \frac{1 + \cos 4t}{8} \right)$$

Allow candidates to start with the given answer and work backwards using the same rules.

So expect to see  $\cos 4t = \pm 1 \pm 2 \sin^2 2t$  or  $\cos 4t = \pm 2 \cos^2 2t \pm 1$  before double angle identities for  $\sin 2t$  or  $\cos 2t$  are used.

**A1\*:** Proceeds to the given answer with correct working. The order of the terms is not important.

Ignore limits for this mark. The integration sign and the  $dt$  must be seen on their final answer.

If they have worked backwards there must be a concluding statement to the effect that they know that they have shown it. The integration sign and the  $dt$  must also be seen

E.g. Reaches  $\int 48 \sin^2 t \cos t + 64 \sin^2 t \cos^2 t \, dt$

$$\begin{aligned}
 \text{Answer is } & \int 8 - 8 \cos 4t + 48 \sin^2 t \cos t \, dt \\
 &= \int 8 - 8(1 - 2 \sin^2 2t) + 48 \sin^2 t \cos t \, dt \\
 &= \int 16 \sin^2 2t + 48 \sin^2 t \cos t \, dt \\
 &= \int 64 \sin^2 t \cos^2 t + 48 \sin^2 t \cos t \, dt \quad \text{which is the same, } \checkmark
 \end{aligned}$$

**B1:** Deduces  $a = \frac{\pi}{4}$ . It may be awarded from the upper limit and can be awarded from (b)

(b)

**M1:** For the key process in using a correct approach to integrating the trigonometric terms.

May be done separately.

There may be lots of intermediate steps (e.g. let  $u = \sin t$ ).

There are other more complicated methods so look carefully at what they are doing.

$$\int 8 - 8 \cos 4t + 48 \sin^2 t \cos t \, dt = \dots \pm P \sin 4t \pm Q \sin^3 t \text{ where } P \text{ and } Q \text{ are constants}$$

**A1:**  $\int 8 - 8 \cos 4t + 48 \sin^2 t \cos t \, dt = 8t - 2 \sin 4t + 16 \sin^3 t (+c)$

If they have written  $16 \sin^3 t$  as  $16 \sin^3$  only award if further work implies a correct answer.

Similarly,  $8t$  may be written as  $8x$ . Award if further work implies  $8t$ , e.g. substituting in their limits.

Do not penalise this sort of slip at all, these are intermediate answers.

**M1:** Uses the limits their  $a$  and 0 where  $a = \frac{\pi}{6}, \frac{\pi}{4}$  or  $\frac{\pi}{3}$  in an expression of the form  $kt \pm P \sin 4t \pm Q \sin^3 t$  leading to an exact answer. Ignore evidence at lower limit as terms are 0

**A1:** CSO  $2\pi + 4\sqrt{2}$  or exact simplified equivalent such as  $2\pi + \frac{8}{\sqrt{2}}$  or  $2\pi + \sqrt{32}$ .

Be aware that  $\int_0^{\frac{\pi}{4}} 8 - 8 \cos 4t + 48 \sin^2 t \cos t \, dt = 8t + \lambda \sin 4t + 16 \sin^3 t (+c)$  would lead to the correct answer but would only score M1 A0 M1 A0

Q12.

| Question   | Scheme                                                                                                                             | Marks | AOs  |
|------------|------------------------------------------------------------------------------------------------------------------------------------|-------|------|
| (a)        | Uses the model to state $\frac{dV}{dt} = -c$ (for positive constant $c$ )                                                          | B1    | 3.1b |
|            | Uses $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$ with their $\frac{dV}{dt} = -c$ and $\frac{dV}{dr} = 4\pi r^2$           | M1    | 2.1  |
|            | $-c = 4\pi r^2 \times \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = -\frac{c}{4\pi r^2} = -\frac{k}{r^2}$ *                            | A1*   | 2.2a |
|            |                                                                                                                                    | (3)   |      |
| (b)        | $\frac{dr}{dt} = -\frac{k}{r^2} \Rightarrow \int r^2 dr = \int -k dt$ and integrates with one side "correct"                       | M1    | 2.1  |
|            | $\frac{r^3}{3} = -kt (+\alpha)$                                                                                                    | A1    | 1.1b |
|            | Uses $t = 0, r = 40 \Rightarrow \alpha = \dots$ $\alpha = \frac{64000}{3}$                                                         | M1    | 1.1b |
|            | Uses $t = 5, r = 20$ & $\alpha = \dots \Rightarrow k = \dots$                                                                      | M1    | 3.4  |
|            | $r^3 = 64000 - 11200t$ or exact equivalent                                                                                         | A1    | 3.3  |
|            |                                                                                                                                    | (5)   |      |
| (c)        | Uses the equation of their model and proceeds to a limiting value for $t$<br>E.g. " $64000 - 11200t$ " ... $0 \Rightarrow t \dots$ | M1    | 3.4  |
|            | For times up to and including $\frac{40}{7}$ seconds                                                                               | A1ft  | 3.5b |
|            |                                                                                                                                    | (2)   |      |
| (10 marks) |                                                                                                                                    |       |      |
| Notes:     |                                                                                                                                    |       |      |





(a)

**B1:** Uses the model to state  $\frac{dV}{dt} = -c$  (for positive constant  $c$ ).

Any "letter" is acceptable here including  $k$ .

Note that  $\frac{dV}{dt} = c$  is B0 unless they state that  $c$  is a negative constant.

**M1:** For an attempt to use  $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$  with their  $\frac{dV}{dt}$  and  $\frac{dV}{dr} = 4\pi r^2$

Allow for an attempt to use  $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$  with their  $\frac{dV}{dt}$  and  $\frac{dV}{dr} = \lambda r^2$  (Any constant is fine)

There is no requirement to use the correct formula for the volume of a sphere for this mark.

**A1\*:** Proceeds to the given answer with an intermediate line equivalent to  $\frac{dr}{dt} = -\frac{c}{4\pi r^2}$

If candidate started with  $\frac{dV}{dt} = -k$  they must provide a minimal explanation how

$$\frac{dr}{dt} = -\frac{k}{4\pi r^2} \rightarrow \frac{dr}{dt} = -\frac{k}{r^2}. \text{ E.g. } \frac{1}{4\pi} \text{ is a constant so replace } \frac{k}{4\pi} \text{ with } k$$

It is not necessary to use the full formula for the volume of a sphere, eg allow  $V = \kappa r^3$  but if it has been quoted it must be correct. So using  $V = 4r^3$  can potentially score 2 of the 3 marks.

(b)

**M1:** For the key step of separating the variables correctly AND integrating one side with at least one index correct. The integral signs do not need to be seen.

**A1:** Correct integration E.g.  $\frac{r^3}{3} = -kt(+\alpha)$  or equivalent. The  $+\alpha$  is not required for this mark.

This may be awarded if  $k$  has been given a value.

**M1:** Uses the initial conditions to find a value for the constant of integration  $\alpha$

If a constant of integration is not present, or  $k$  has been given a pre defined value, then only the first two marks can be awarded in part (b)

The mark may be awarded if the equation has been adapted incorrectly. E.g. each term cube rooted.

**M1:** Uses the second set of conditions with their value of  $\alpha$  to find  $k$

This may be awarded if the equation has been adapted incorrectly. E.g. each term cube rooted.

**A1:** Obtains any correct equation for the model.

$$\text{E.g. } r^3 = 64000 - 11200t \text{ or exact equivalent such as } \frac{r^3}{3} = \frac{64000}{3} - \frac{11200}{3}t.$$

ISW after sight of a correct answer. Condone recurring decimals e.g. 21333.3 for  $\frac{64000}{3}$

Do not award if only the rounded/truncated decimal equivalents to say  $\frac{64000}{3}$  is used.

(c)

**M1:** Recognises that the model is only valid when  $r \geq 0$  and uses this to find  $t$ . Condone  $r > 0$

Award for an attempt to find the value of  $t$  when  $r = 0$ . See scheme.

It must be from an equation of the form  $ar^n = b - ct$ ,  $a, b, c > 0$  which give +ve values of  $t$ .

**A1ft:** Allow valid for times up to (and including)  $\frac{40}{7}$  seconds, 5.71 seconds. Allow  $t < \frac{40}{7}$  or  $t \leq \frac{40}{7}$

There is no requirement for the left hand side of the inequality, 0

States invalid for times greater than  $\frac{40}{7}$  seconds, 5.71 seconds.

Follow through on their equation so allow  $t < \text{their } \frac{64000}{11200}$  as long as this value is greater than 5

( $t = 5$  is one of the values in the question)



Q13.

| Question   | Scheme                                                                                                                                                              | Marks    | AOs          |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------|
|            | $\frac{dy}{dx} = \frac{15}{2}x^{\frac{1}{2}} - 9$                                                                                                                   | M1<br>A1 | 3.1a<br>1.1b |
|            | Substitutes $x = 4 \Rightarrow \frac{dy}{dx} = 6$                                                                                                                   | M1       | 2.1          |
|            | Uses (4, 15) and gradient $\Rightarrow y - 15 = 6(x - 4)$                                                                                                           | M1       | 2.1          |
|            | Equation of $l$ is $y = 6x - 9$                                                                                                                                     | A1       | 1.1b         |
|            | Area $R = \int_0^4 \left( 5x^{\frac{3}{2}} - 9x + 11 \right) - (6x - 9) dx$                                                                                         | M1       | 3.1a         |
|            | $= \left[ 2x^{\frac{5}{2}} - \frac{15}{2}x^2 + 20x + c \right]_0^4$                                                                                                 | A1       | 1.1b         |
|            | Uses both limits of 4 and 0<br>$\left[ 2x^{\frac{5}{2}} - \frac{15}{2}x^2 + 20x \right]_0^4 = 2 \times 4^{\frac{5}{2}} - \frac{15}{2} \times 4^2 + 20 \times 4 - 0$ | M1       | 2.1          |
|            | Area of $R = 24$ *                                                                                                                                                  | A1*      | 1.1b         |
|            | Correct notation with good explanations                                                                                                                             | A1       | 2.5          |
|            |                                                                                                                                                                     | (10)     |              |
| (10 marks) |                                                                                                                                                                     |          |              |

**Notes:**

**M1:** Differentiates  $5x^{\frac{3}{2}} - 9x + 11$  to a form  $Ax^{\frac{1}{2}} + B$

**A1:**  $\frac{dy}{dx} = \frac{15}{2}x^{\frac{1}{2}} - 9$  but may not be simplified

**M1:** Substitutes  $x = 4$  in their  $\frac{dy}{dx}$  to find the gradient of the tangent

**M1:** Uses their gradient and the point (4, 15) to find the equation of the tangent

**A1:** Equation of  $l$  is  $y = 6x - 9$

**M1:** Uses Area  $R = \int_0^4 \left( 5x^{\frac{3}{2}} - 9x + 11 \right) - (6x - 9) dx$  following through on their  $y = 6x - 9$

Look for a form  $Ax^{\frac{5}{2}} + Bx^2 + Cx$

**A1:**  $= \left[ 2x^{\frac{5}{2}} - \frac{15}{2}x^2 + 20x + c \right]_0^4$  This must be correct but may not be simplified

**M1:** Substitutes in both limits and subtracts

**A1\*:** Correct area for  $R = 24$

**A1:** Uses correct notation and produces a well explained and accurate solution. Look for

- Correct notation used consistently and accurately for both differentiation and integration
- Correct explanations in producing the equation of  $l$ . See scheme.
- Correct explanation in finding the area of  $R$ . In way 2 a diagram may be used.

Alternative method for the area using area under curve and triangles. (Way 2)

**M1:** Area under curve  $= \int_0^4 \left( 5x^{\frac{3}{2}} - 9x + 11 \right) = \left[ Ax^{\frac{5}{2}} + Bx^2 + Cx \right]_0^4$

**A1:**  $= \left[ 2x^{\frac{5}{2}} - \frac{9}{2}x^2 + 11x \right]_0^4 = 36$

**M1:** This requires a full method with all triangles found using a correct method

Look for Area  $R = \text{their } 36 - \frac{1}{2} \times 15 \times \left( 4 - \text{their } \frac{3}{2} \right) + \frac{1}{2} \times \text{their } 9 \times \text{their } \frac{3}{2}$

Q14.

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Scheme                                                                                |                                                            | Marks    | AOs          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------|----------|--------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\int (3x^{0.5} + A) dx = 2x^{1.5} + Ax(+c)$                                          |                                                            | M1<br>A1 | 3.1a<br>1.1b |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Uses limits and sets $= 2A^2 \Rightarrow (2 \times 8 + 4A) - (2 \times 1 + A) = 2A^2$ |                                                            | M1       | 1.1b         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Sets up quadratic and attempts to solve                                               | Sets up quadratic and attempts $b^2 - 4ac$                 | M1       | 1.1b         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\Rightarrow A = -2, \frac{7}{2}$ and states that there are two roots                 | States $b^2 - 4ac = 121 > 0$ and hence there are two roots | A1       | 2.4          |
| <b>(5 marks)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                            |          |              |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                       |                                                            |          |              |
| <p><b>M1:</b> Integrates the given function and achieves an answer of the form <math>kx^{1.5} + Ax(+c)</math> where <math>k</math> is a non- zero constant</p> <p><b>A1:</b> Correct answer but may not be simplified</p> <p><b>M1:</b> Substitutes in limits and subtracts. This can only be scored if <math>\int A dx = Ax</math> and not <math>\frac{A^2}{2}</math></p> <p><b>M1:</b> Sets up quadratic equation in <math>A</math> and either attempts to solve or attempts <math>b^2 - 4ac</math></p> <p><b>A1:</b> Either <math>A = -2, \frac{7}{2}</math> and states that there are two roots</p> <p>Or states <math>b^2 - 4ac = 121 &gt; 0</math> and hence there are two roots</p> |                                                                                       |                                                            |          |              |

Q15.

| Question   | Scheme                                                                                                                                                                                             | Marks | AOs  |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|
| (a)        | Uses or implies $h = 0.5$                                                                                                                                                                          | B1    | 1.1b |
|            | For correct form of the trapezium rule =                                                                                                                                                           | M1    | 1.1b |
|            | $\frac{0.5}{2} \{3 + 2.2958 + 2(2.3041 + 1.9242 + 1.9089)\} = 4.393$                                                                                                                               | A1    | 1.1b |
|            |                                                                                                                                                                                                    | (3)   |      |
| (b)        | Any valid statement reason, for example <ul style="list-style-type: none"> <li>• Increase the number of strips</li> <li>• Decrease the width of the strips</li> <li>• Use more trapezia</li> </ul> | B1    | 2.4  |
|            |                                                                                                                                                                                                    | (1)   |      |
| (c)        | For integration by parts on $\int x^2 \ln x \, dx$                                                                                                                                                 | M1    | 2.1  |
|            | $= \frac{x^3}{3} \ln x - \int \frac{x^2}{3} \, dx$                                                                                                                                                 | A1    | 1.1b |
|            | $\int -2x + 5 \, dx = -x^2 + 5x \quad (+c)$                                                                                                                                                        | B1    | 1.1b |
|            | All integration attempted and limits used                                                                                                                                                          |       |      |
|            | Area of $S = \int_1^3 \frac{x^2 \ln x}{3} - 2x + 5 \, dx = \left[ \frac{x^3}{9} \ln x - \frac{x^3}{27} - x^2 + 5x \right]_{x=1}^{x=3}$                                                             | M1    | 2.1  |
|            | Uses correct ln laws, simplifies and writes in required form                                                                                                                                       | M1    | 2.1  |
|            | Area of $S = \frac{28}{27} + \ln 27 \quad (a = 28, b = 27, c = 27)$                                                                                                                                | A1    | 1.1b |
|            |                                                                                                                                                                                                    | (6)   |      |
| (10 marks) |                                                                                                                                                                                                    |       |      |

**Notes:**

(a)

**B1:** States or uses the strip width  $h = 0.5$ . This can be implied by the sight of  $\frac{0.5}{2}\{\dots\}$  in the trapezium rule

**M1:** For the correct form of the bracket in the trapezium rule. Must be  $y$  values rather than  $x$  values  $\{\text{first } y \text{ value} + \text{last } y \text{ value} + 2 \times (\text{sum of other } y \text{ values})\}$

**A1:** 4.393

(b)

**B1:** See scheme

(c)

**M1:** Uses integration by parts the right way around.

Look for  $\int x^2 \ln x \, dx = Ax^3 \ln x - \int Bx^2 \, dx$

**A1:**  $\frac{x^3}{3} \ln x - \int \frac{x^2}{3} \, dx$

**B1:** Integrates the  $-2x + 5$  term correctly  $= -x^2 + 5x$

**M1:** All integration completed and limits used

**M1:** Simplifies using  $\ln$  law(s) to a form  $\frac{a}{b} + \ln c$

**A1:** Correct answer only  $\frac{28}{27} + \ln 27$